

Coalescence-Induced Droplet Jumping on Superhydrophobic Surfaces with Annular Wedge-Shaped Micropillar Arrays

Published as part of Langmuir *virtual special issue* “Highlighting Contributions from our Editorial Board Members in 2023”.

Huimin Hou, Xiaomin Wu,* Zhifeng Hu, Sihang Gao, and Zhiping Yuan*



Cite This: *Langmuir* 2023, 39, 18825–18833



Read Online

ACCESS |



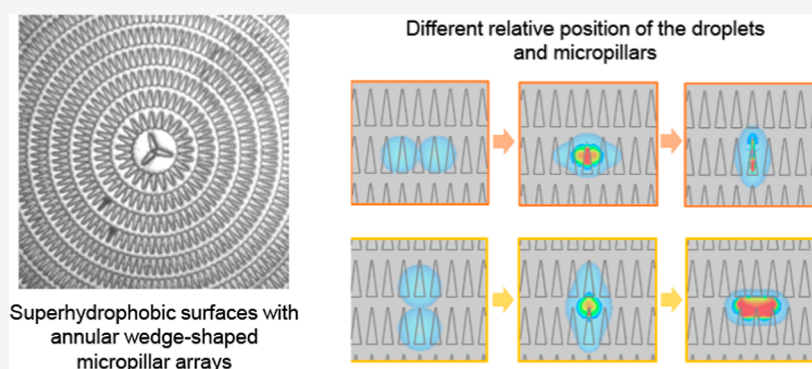
Metrics & More



Article Recommendations



Supporting Information



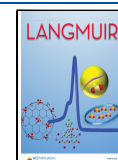
ABSTRACT: The coalescence-induced droplet jumping on superhydrophobic surfaces has extensive application potential in water harvesting, thermal management of electronic devices, and microfluidics. The rational design of the surface structure can influence the interaction between the droplet and the surface, thereby controlling the velocity and direction of the droplet's jumping. In this study, we fabricate the superhydrophobic surface with annular wedge-shaped micropillar arrays, examine the dynamic behavior of condensate droplets on the surface, and measure the temporal and spatial variations of droplet density, average radius, and surface coverage with wedge-shaped micropillars of varying sizes. In addition, the energy analysis of the coalescence-induced droplet jumping reveals that the two primary factors influencing the jumping are the relative size and position of the droplets and micropillars. Further numerical simulations find that the wedge-shaped micropillars cause an asymmetric distribution of pressure within the droplet and at the solid–liquid contact surface, which generates an unbalanced force driving the droplet in the gradient direction of the wedge-shaped micropillar, causing the droplet to jump off the surface with both vertical and gradient-direction velocities. The capacity of the wedge-shaped micropillar surface to transport droplets in the gradient direction increases and then decreases as the relative size of the droplets and micropillars increases. The relative position of the droplet center-of-mass line perpendicular to the bottom edge of the wedge micropillars' trapezoidal shape is more favorable for droplet transport. This work reveals the influence mechanism of surface structure on the velocity and direction of droplet jumping, and the results can guide the microstructure design of superhydrophobic surfaces, which has significant implications for the application of droplet jumping.

INTRODUCTION

Coalescence-induced droplet jumping on superhydrophobic surfaces has attracted much attention due to its great potential for applications in enhanced condensation,^{1–3} water harvesting,^{4,5} thermal management of electronic devices,^{6–8} and microfluidics.^{9–11} In 2009, Boreyko et al.¹² discovered the droplet jumping phenomenon on superhydrophobic surfaces with carbon nanotubes and silicon micropillars, measured the jumping velocities of droplets with radii ranging from 8 to 160 μm , and proposed that the jumping velocities follow the inertia-capillary scaling law. Coalescence-induced droplet jumping is triggered by the interaction between the droplet and the surface and the conversion of a portion of the released

surface energy during droplet coalescence into the kinetic energy of jumping. The rational design of the surface structure can influence the liquid's flow characteristics in the coalescing process and the droplet's interaction with the surface to control the velocity and direction of the droplet jumping.^{13–18} Vahabi

Received: August 29, 2023
Revised: October 23, 2023
Accepted: November 27, 2023
Published: December 14, 2023



et al.,¹⁹ for instance, designed a superhydrophobic surface with a ridge-like raised structure to increase the energy conversion rate by approximately 570% compared to droplet jumping on a superhydrophobic horizontal surface, accompanied by a significant increase in jumping velocity and height. Yuan et al.²⁰ designed the sloped surface structure to control the direction of droplet jumping. Lu et al.²¹ used the asymmetric V-groove to obtain jumping velocities of up to $0.61v_{ic}$ (the capillary-inertial velocity), and by adjusting the angle of the V-groove, they caused the jumping direction to deviate from the base surface by up to 60° . In addition to superhydrophobic surfaces with individual structures, experiments and numerical simulations have also been used to study the droplet motion on surfaces with micropillar arrays.^{22–26} The four micropillar configurations investigated more are cylindrical, rectangular, conical, and wedge-shaped. Mulore et al.²⁷ conducted condensation experiments on the superhydrophobic surface with nanocylinders and discovered that nanopillar height, diameter, and spacing significantly impacted the critical jump diameter of droplets. Tall and skinny nanocylinders could facilitate the jumping of $2\ \mu\text{m}$ droplets and reduce the critical jump diameter by 1 order of magnitude. Attarzadeh and Dolatabadi²⁸ used numerical simulation to study droplet jumping on the superhydrophobic surface with rectangular micropillars. They discovered that the width and spacing of the micropillars affected the droplet volume immersion into the spaces between the micropillars, which in turn affected the droplet jump velocity. The greater the droplet's immersion of the gaps, the greater the viscous dissipation and the lower the jump velocity. The numerical results of Shi et al.²⁹ for droplet jumping on the superhydrophobic surface with conical micropillars demonstrated that the jump height increased with the increase in the height of the conical micropillars and the decrease in their spacing. For wedge-shaped micropillars, Wang et al.⁵ reported tilted droplet jumping and remote transportation on the hybrid wetting surface with wedge-shaped patterns, significantly increasing the surface water-harvesting capacity. Liu et al.³⁰ designed the microanisotropic superhydrophobic surface, which consists of successive periodic isosceles triangle microarrays and could cause spontaneous jumping of condensate droplets in the prescribed direction relying on the three-phase linear imbalance force. After jumping off the superhydrophobic surface with wedge-shaped micropillar arrays, the droplets could be spatially transported, which facilitates the application of droplet jumping in water harvesting, thermal management of electronic devices, and microfluidics.

However, the investigation of coalescence-induced droplet jumping on superhydrophobic surfaces with wedge-shaped micropillars is insufficient. First, the temporal and spatial variations in the number and radius of condensate droplets on the wedge-shaped micropillar surface resulting from the dynamic behavior of the droplets are unknown. Most wedge-shaped micropillars in existing studies are arranged in linear arrays. The distribution of droplets in each area is similar on such surfaces, and obtaining the spatial changes in the quantity and radius of droplets is difficult. In addition, it is necessary to elucidate the factors that affect droplet jumping on the wedge-shaped micropillar surface and their influence on the velocity and direction of droplet jumping. In this work, we design the superhydrophobic surface with annular wedge-shaped micropillar arrays, observe the movement of condensate droplets, and statistically obtain the variation of droplet number density,

average droplet radius over time, and droplet spatial distribution on the surface. Regarding energy, the coalescence-induced droplet jumping on the wedge-shaped micropillar is also analyzed, and the effects of the relative size and position of the droplets and micropillars are further elucidated through numerical simulation. The findings of this study can provide theoretical guidance for the applicability of droplet jumping in various fields.

MATERIALS AND EXPERIMENTS

Surface Fabrication. To obtain the droplet distribution in different areas and explore the interaction between wedge-shaped micropillars and droplets, a mask plate with an array of annular wedge-shaped micropillars was designed, and a specialized unit (Institute of Superlubricity Technology, Research Institute of Tsinghua University in Shenzhen) was commissioned to fabricate a silicon surface with wedge-shaped micropillars by using standard photolithography and etching processes. The surface structure is depicted in Figure 1a. C_i is the number of rings of the wedge-shaped

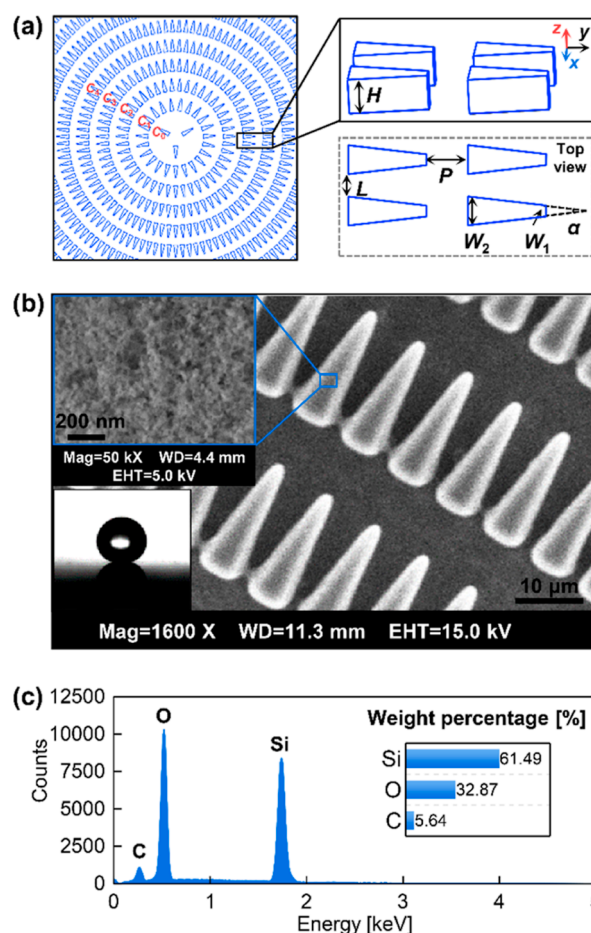


Figure 1. Morphology of the superhydrophobic surface with annular wedge-shaped micropillar arrays. (a) Schematic diagram of the surface structure; (b) SEM image of the surface and the contact angle of the droplet on the surface; (c) elemental composition of the surface.

micropillar counting from the center of the surface. H is the height of the wedge-shaped micropillar, α is the angle of the wedge-shaped micropillar on both sides, P and L are the ring spacing and micropillar spacing, respectively, and W_1 and W_2 are the top and bottom edges of the wedge-shaped micropillar, respectively. The photolithographically processed samples were sequentially rinsed in acetone, ethanol, and deionized water; dried under a nitrogen flow; and then immersed for

5 min in a commercial superhydrophobic coating solution (Glaco Mirror Coat Zero, Soft 99, Japan). After drying vertically in air, the samples were placed into an oven heated to 120 °C for 15 min, and the above infiltration–drying procedure was repeated three times. The surface morphology of the samples was examined by using a scanning electron microscope (FE-SEM, Zeiss Merlin, Germany). As shown in Figure 1b, wedge-shaped micropillars with regular morphologies were visible on the surface, and a layer of nanoparticles covered each micropillar uniformly. The surface's elemental composition was analyzed by using an energy spectrometer (EDS, Zeiss Merlin, Germany). As shown in Figure 1c, the primary surface elements were Si, C, and O. The static contact angle, advancing angle, receding angle, and rolling angle of a 3 μ L deionized water droplet on the surface were measured by a contact angle measuring instrument (JC2000C1), which were 162.3 ± 1.7 , 165.1 ± 1.3 , 160.8 ± 1.9 , $2.5 \pm 1.0^\circ$, respectively. In this work, three kinds of surfaces with wedge-shaped micropillars were considered, named T-2.5, T-5, and T-10. The bottom edge W_2 of the wedge-shaped micropillars on each of the three surfaces was 2.5, 5, 10 μ m, while the micropillar spacing L was 3.1, 3.3, 3.8 μ m. However, the ring spacing P (10 μ m), the top edge W_1 of the wedge-shaped micropillar (1 μ m), the angle α of the wedge-shaped micropillar on both sides (15°), the micropillar height H (10 μ m), and the number of micropillar rings C (300) were the same on the three surfaces.

Experimental Apparatus. Figure 2 depicts the condensation apparatus for observing droplet motion. The experimental setup

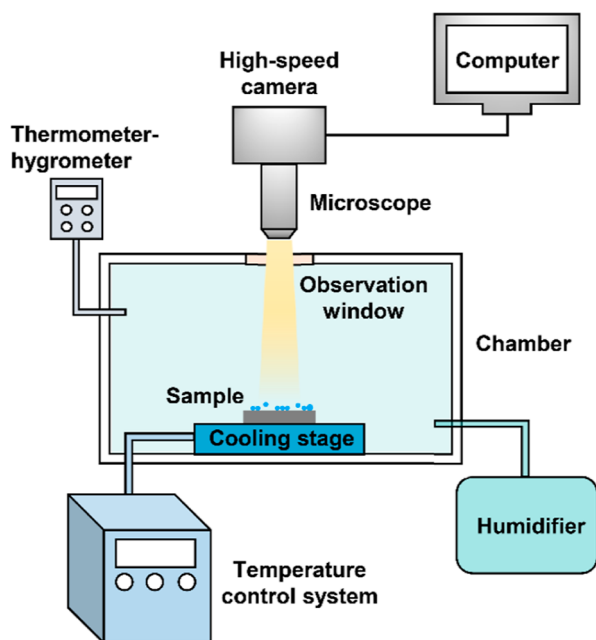


Figure 2. Schematic diagram of the condensation experiment setup.

mainly consisted of a chamber, a thermometer–hygrometer, a high-speed camera, a computer, a humidifier, and a cooling stage. The poly(methyl methacrylate) plate was used to construct the chamber to ensure constant temperature and humidity in the condensation environment, and an observation window was set above the chamber to observe condensation droplets' dynamic behavior. The air generated by the humidifier containing many microdroplets was heated by an electric filament and turned into water vapor, which then entered the condensation chamber. The ambient temperature in the experiment was 26 ± 1 °C, and the relative humidity was $94 \pm 5\%$, as measured by the thermometer–hygrometer (DS-NT2, Rotronic Hygrolog). The superhydrophobic surface with wedge-shaped micropillars was attached to the cooling stage with thermally conductive silicone grease, and the surface temperature was maintained at 15 ± 1 °C. A high-speed camera (CP80-3-M-540, Optronis) combined with a microscope (AFT-0750X-20-C, Micro-

vision) was used to record the droplet motion on the surface from the top view at 50 fps.

RESULTS AND DISCUSSION

Dynamic Behavior of Condensate Droplets. Figure 3 depicts three dynamic behaviors of condensate droplets on the

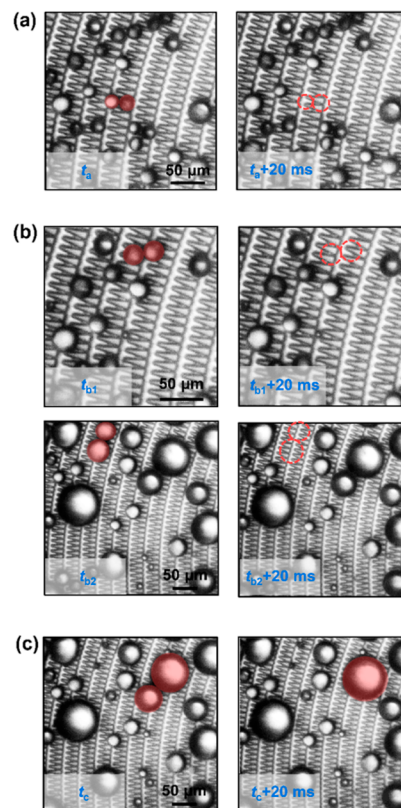


Figure 3. Dynamic behavior of condensate droplets on a T-10 superhydrophobic surface with annular wedge-shaped micropillar arrays. (a) Coalescence-induced droplet jumping on a single wedge-shaped micropillar ($t_a = 213.7$ s); (b) coalescence-induced droplet jumping across wedge-shaped micropillars ($t_{b1} = 234.66$ s, $t_{b2} = 543.3$ s); (c) droplet coalescence without significant movement ($t_c = 576.06$ s).

superhydrophobic surface with annular wedge-shaped micropillar arrays: coalescence-induced droplet jumping on a single wedge-shaped micropillar, coalescence-induced droplet jumping across wedge-shaped micropillars, and droplet coalescence without significant movement. During the condensation process, the growing condensate droplet on a single wedge-shaped micropillar coalesced with nearby droplets of similar size, and a portion of the surface energy released was converted into jumping kinetic energy. Therefore, the droplets departed the surface with a particular velocity. The dimension of a single droplet was smaller than that of a single wedge-shaped micropillar at this time (Figure 3a). If there were no other droplets of comparable size nearby, the droplet would continue to grow to a size larger than that of a single wedge-shaped micropillar. At that time, the droplet spanned multiple micropillars and could also leave the surface via coalescence-induced jumping after contacting a droplet of comparable size (Figure 3b). In addition, some droplets did not endure significant movement after coalescence because the released surface energy was not converted to sufficient kinetic energy to

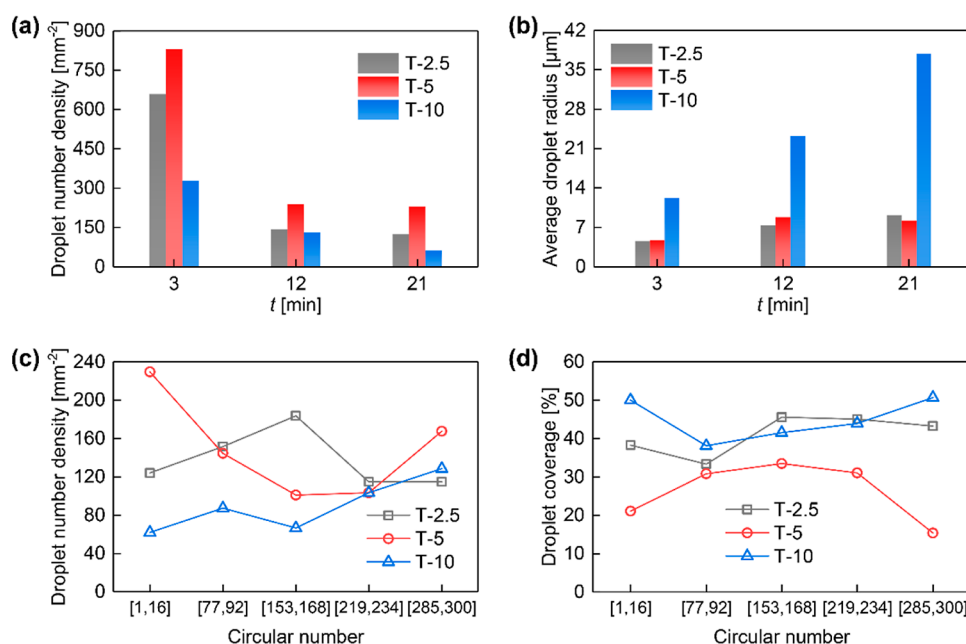


Figure 4. On the T-2.5, T-5, and T-10 superhydrophobic surfaces with annular wedge-shaped micropillar arrays, (a,b) the variation of droplet number density and average radius with time in the center region of three surfaces. The three surfaces' central regions all had an area of $0.66 \times 0.66 \text{ mm}^2$ and contained wedge-shaped micropillars of the first to 16th rings; (c,d) the variation of droplet number density and droplet coverage of the different regions of the surfaces at 21 min of condensation.

propel the droplets to jump off the surface (Figure 3c). The detailed condensation process is depicted in Supporting Information Video S1. On superhydrophobic surfaces with annular wedge-shaped micropillar arrays, coalescence-induced droplet jumping on a single micropillar and across micropillars were the predominant modes of droplet departure from the surface and the predominant behaviors that altered the droplet distribution on the surface.

To demonstrate the effect of the dynamic behavior of droplets on the droplet distribution on superhydrophobic surfaces with different sizes of wedge-shaped micropillar arrays, the variation of droplet number density and average radius with time in the center region of three surfaces, T-2.5, T-5, and T-10, was statistically investigated, as shown in Figure 4a,b. The three surfaces' central regions all had an area of $0.66 \times 0.66 \text{ mm}^2$ and contained wedge-shaped micropillars of the first to 16th rings. As shown in Figure 4a, as condensation time increased, the droplet number density on the three surfaces decreased gradually, because the number of droplets decreased due to droplet coalescence and jumped off the surface. However, as shown in Figure 4b, the average radius of the droplet progressively increased, which may be because the droplet jumped off the surface, then fell back, and coalesced with the surface droplets. The radius of the droplets increased as the merged droplets continued to grow or coalesce with the surrounding droplets. The T-10 surface produced droplets with a maximal size of approximately $132 \mu\text{m}$ after 21 min of condensation, which was significantly larger than the wedge-shaped micropillar. A comparison of the average radius of droplets on the three surfaces revealed that T-2.5 had the smallest droplet radius and T-10 had the largest. When wedge-shaped micropillars were present on the surface, droplets grew, coalesced, and jumped on the micropillars, and the size of the wedge-shaped micropillars affected the size of the droplets. A comparison of Figure 4a,b revealed that the droplet number density on the T-5 surface was the greatest, whereas the

average radius was comparable to that of T-2.5. The reason may be that the frequency of the droplet jumping on the T-5 surface was greater, and after the droplet jumped away from the surface, the original position was renucleated and tiny droplets grew, resulting in a greater droplet number density but a smaller average radius. In addition, Figure 4c,d displayed the droplet number density and coverage in five regions from the surface's center outward, indicating that the wedge-shaped micropillars' size influenced the droplets' spatial distribution. On the T-2.5 surface, droplet number density and coverage were greatest in the middle region. In contrast, on the T-5 surface, the droplet number density decreased and then increased from the surface's center to its periphery, whereas droplet coverage varied in the opposite direction. On the T-10 surface, the droplet number density increased progressively from the surface's center to its periphery, whereas droplet coverage decreased and then increased.

Energy Analysis of Coalescence-Induced Droplet Jumping. The energy analysis of the coalescence-induced jumping was performed to elucidate the mechanism of droplet jumping on a superhydrophobic surface with wedge-shaped micropillars. The kinetic energy obtained by the droplet when it jumped off the surface is^{31,32}

$$E_k = \Delta E_s - E_w - E_{vis} - E_h \quad (1)$$

where ΔE_s , E_w , E_{vis} , and E_h represent the surface energy released by droplet coalescence, the adhesion work, the viscous dissipation, and the increased gravitation potential energy, respectively. Since droplet diameter is smaller than capillary length ($\sqrt{\sigma/\rho g} = 2.7 \text{ mm}$), E_h is neglected.³³ To facilitate calculations, it is supposed that two equal-sized droplets coalesce and jump, and the surface energy released by droplet coalescence is calculated as follows

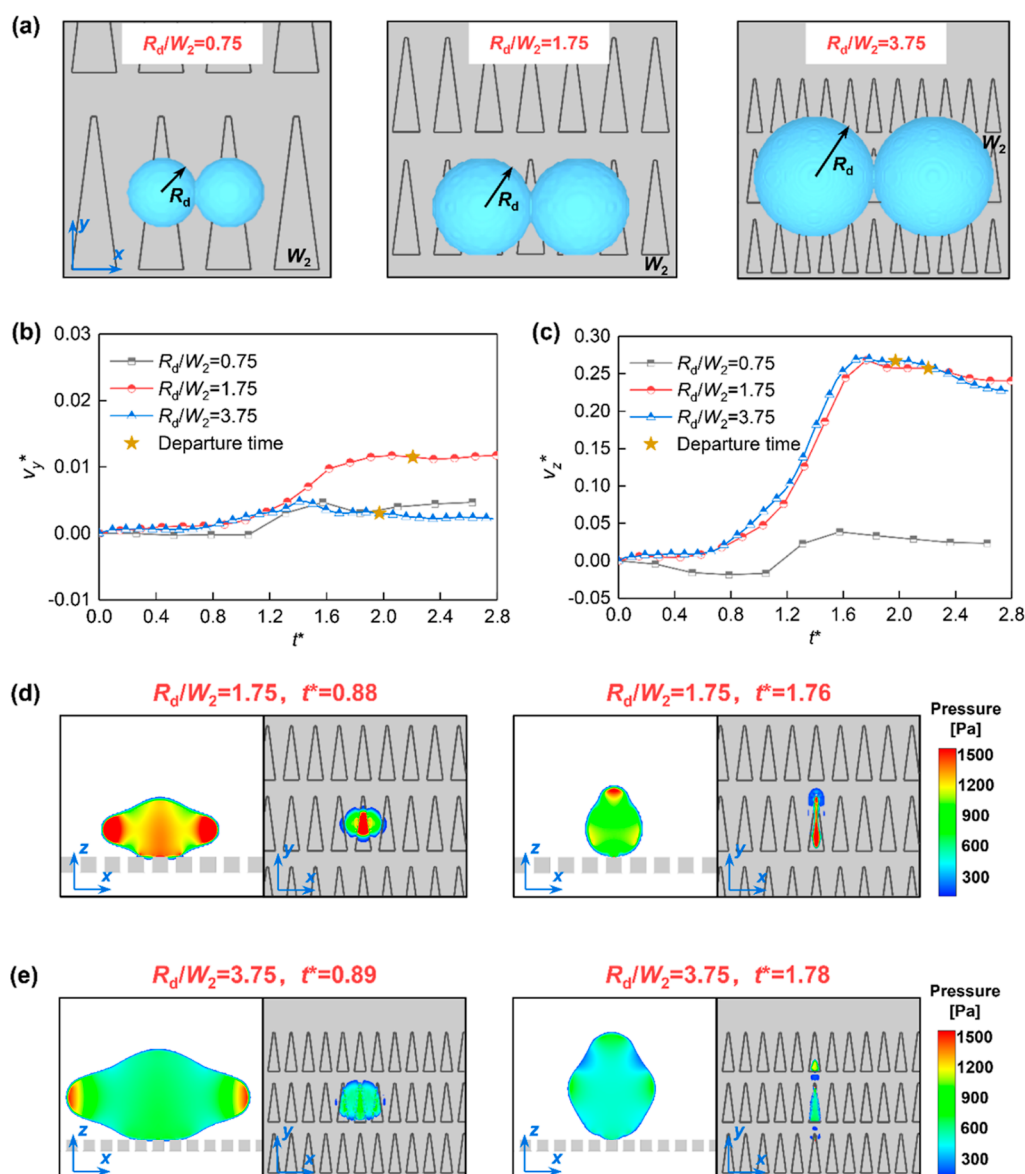


Figure 5. Effect of the relative size of droplets and micropillars on coalescence-induced droplet jumping when the droplet center-of-mass line on the T-5 surface is parallel to the trapezoidal bottom edge of the wedge-shaped micropillar. (a) The initial state of the top view of different droplet-micropillar relative sizes; (b,c) the variation of dimensionless gradient-direction velocity v_y^* and vertical velocity v_z^* of the droplet center of mass with time for different droplet-micropillar relative sizes; (d,e) the pressure distribution inside the droplet and at the solid–liquid contact surface for $R_d/W_2 = 1.75$ and 3.75 .

$$\Delta E_s = 4\pi R_d \sigma_{lv} (1 - \cos \theta) - \sigma_{lv} \cos \theta (A_{sl,1} + A_{sl,2}) - 4\pi \sigma_{lv} R_e^2 \quad (2)$$

The adhesion work is

$$E_w = (A_{sl,1} + A_{sl,2}) \sigma_{lv} (1 + \cos \theta) \quad (3)$$

The viscous dissipation is⁵

$$E_{vis} = 72\pi\mu \sqrt{\frac{\sigma_{lv} R_d^3}{\rho}} \quad (4)$$

where R_d is the initial droplet radius, σ_{lv} is the gas–liquid surface tension, θ is the contact angle, $A_{sl,1}$ and $A_{sl,2}$ are the contact areas of the two droplets with the surface, respectively, R_e is the equivalent radius after the coalescence of the two droplets, and μ and ρ are the viscosity and density of the

droplets, respectively. See Section S1 of the Supporting Information for the detailed derivation process of eqs 2 and 3 and the calculation of the solid–liquid contact area. From eqs 2–4, it can be determined that when the structural parameters of wedge-shaped micropillars and the types of droplets are constant, the released surface energy and adhesion work are primarily influenced by R_d , $A_{sl,1}$, and $A_{sl,2}$, and the viscous dissipation is primarily influenced by R_d . Combined with eq 1, we can observe that the kinetic energy of the droplet when it jumps off the surface is primarily influenced by R_d , $A_{sl,1}$, and $A_{sl,2}$. Therefore, the relative size and relative position of the droplets and micropillars are the primary factors influencing coalescence-induced droplet jumping on the surface with wedge-shaped micropillars.

Effect of the Relative Size and Position of the Droplets and Micropillars. To further investigate the influence of the relative size and position of the droplets and

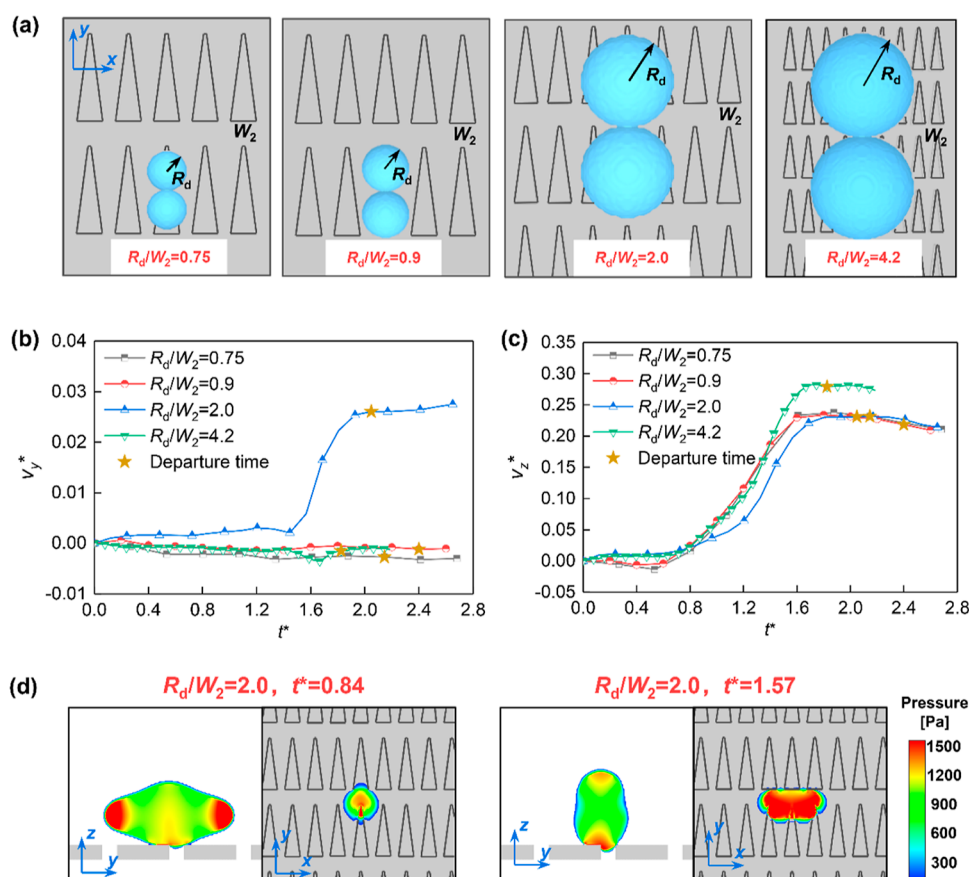


Figure 6. Effect of the relative size of droplets and micropillars on coalescence-induced droplet jumping when the droplet center-of-mass line on the T-S surface is perpendicular to the trapezoidal bottom edge of the wedge-shaped micropillar. (a) The initial state of the top view of different droplet-micropillar relative sizes; (b,c) the variation of dimensionless gradient-direction velocity v_y^* and vertical velocity v_z^* of the droplet center of mass with time for different droplet-micropillar relative sizes; (d) the pressure distribution inside the droplet and at the solid–liquid contact surface for $R_d/W_2 = 2.0$.

micropillars on droplet jumping, interFoam,^{20,34} a multiphase flow solver based on the VOF (volume of fluids) method in OpenFOAM software, was used to simulate the coalescence-induced droplet jumping process on the surface with annular wedge-shaped micropillar arrays. Experiments found that the relative positions of the droplets and micropillars mainly included two kinds: the droplet center-of-mass line was parallel (i.e., along the x -direction) and perpendicular (i.e., along the y -direction) to the trapezoidal bottom edge of the wedge-shaped micropillar. The number of wedge-shaped micropillars in contact with the droplet during the coalescence-induced jumping was small, and for the convenience of calculation, the annular array of wedge-shaped micropillars was regarded as a linear array in the numerical simulation (Section S2, Supporting Information). Consistent with previous research,^{35,36} the droplet velocity v and time t were dimensionless with the capillary-inertial velocity v_{ic} ($=\sqrt{\sigma_{lv}/\rho R_e}$) and the capillary-inertial time t_{ic} ($=\sqrt{\rho R_e^3/\sigma_{lv}}$), respectively: $v^* = v/v_{ic}$, $t^* = t/t_{ic}$. The dimensionless velocity and time were used to characterize the droplets' coalescence and jumping behavior on surfaces with wedge-shaped micropillars.

Figure 5a depicted the initial state of the top view of different relative sizes (R_d/W_2) of the droplets and micropillars when the droplet center-of-mass line was parallel to the trapezoidal bottom edge of the wedge-shaped micropillar, and

the larger the droplet-micropillar relative size, the more the droplet touched the micropillar. The results of numerical simulations indicated that the velocity of the droplet center of mass in the x -direction was much smaller than that in the y - and z -directions so that it was disregarded, and only the change of the dimensionless gradient-direction velocity v_y^* and the vertical velocity v_z^* of the droplet center of mass with time were taken into account. As depicted in Figure 5b,c, when the droplet radius was small ($R_d/W_2 = 0.75$), the gradient-direction velocity and vertical velocity were both small and the droplet failed to jump off the surface due to excessive viscous dissipation and insufficient conversion of surface energy to kinetic energy. With an increase in the relative size of the droplets and micropillars, the gradient-direction velocity decreased when the droplet jumped off the surface, the vertical velocity did not vary significantly, and the departure time was accelerated. The pressure distribution inside the droplet and at the solid–liquid contact surface when $R_d/W_2 = 1.75$ was depicted in Figure 5d, which revealed that, on the surface with wedge-shaped micropillars, the droplet expanded along the micropillars and their gap after making contact with the micropillar. The droplet's internal pressure and solid–liquid contact surface pressure were symmetrically distributed along the x -direction. In contrast, the pressure along the y -direction was distributed asymmetrically, resulting in an unbalanced force driving the droplet's motion in the y -direction and making the droplet jump from the trapezoidal

bottom edge of the micropillar to the top edge. The relative transport distance L_t/R_d of the droplet in the gradient direction was defined as the ratio of the transport distance to the initial droplet radius when air resistance was neglected^{4,34}

$$\frac{L_t}{R_d} = \frac{2v_{y,j}v_{z,j}}{R_d g} \quad (5)$$

where $v_{y,j}$ and $v_{z,j}$ are the gradient-direction velocity and vertical velocity of the droplet center of mass at the departure time. The relative transport distance L_t/R_d was calculated to be 4.5 when $R_d/W_2 = 1.75$, substantially greater than the relative transport distance of 0.3 when $R_d/W_2 = 3.75$. As shown in Figure 5d,e, although the contact surface between the droplets and the micropillars was greater when the droplets were larger, the asymmetry of the pressure distribution at the solid–liquid contact surface was diminished and the droplets' ability to move in the gradient direction decreased. Furthermore, when there were no wedge-shaped micropillars on the superhydrophobic surface, the pressure inside the droplet and at the solid–liquid contact surface was symmetrically distributed in both the x and y directions. The droplet jumped away from the surface with a horizontal velocity close to zero and could not be transported horizontally (Section S3, Supporting Information). Overall, the wedge-shaped micropillar superhydrophobic surface caused the droplets to have vertical and gradient-direction velocities when they jumped off the surface. When the droplet center-of-mass line was parallel to the trapezoidal bottom edge of the wedge-shaped micropillar, the gradient-direction droplet transport capacity of the wedge-shaped micropillar surface first increased and then decreased as the relative size of the droplet-micropillar increased. The droplet transport capacity could be regulated by rationally designing the relative size of the droplets and micropillars.

Figure 6a depicts the initial state of the top view of different relative sizes (R_d/W_2) of the droplets and micropillars when the droplet center-of-mass line is perpendicular to the trapezoidal bottom edge of the wedge-shaped micropillar. With increasing droplet-micropillar relative size, there was a shift from droplet jumping on a single wedge-shaped micropillar to droplet jumping across wedge-shaped micropillars. The variation of dimensionless gradient-direction velocity v_y^* and vertical velocity v_z^* of the droplet center of mass were depicted in Figure 6b,c, from which it could be observed that, as the relative size of the droplet-micropillar increased, the time when the droplet left the surface was first delayed and then advanced, and the vertical velocity of the droplet increased. When the relative size of the droplet-micropillar was smaller ($R_d/W_2 = 0.75$ and 0.9) or larger ($R_d/W_2 = 4.2$), the gradient-direction velocity approached zero, and the wedge-shaped micropillar could not propel the droplet in the gradient direction. The pressure distribution inside the droplet and at the solid–liquid contact surface when $R_d/W_2 = 2.0$ is shown in Figure 6d. Similar to Figure 5d, the presence of the wedge-shaped micropillar caused the pressure inside the droplet and at the solid–liquid contact surface to be symmetrically distributed along the x -direction and asymmetrically distributed along the y -direction, thereby generating an unbalanced force in the y -direction that drove the droplet's movement. Comparing Figures 5 and 6, it was evident that compared with the droplet center-of-mass line parallel to the trapezoidal bottom edge of the wedge-shaped micropillar, in the perpendicular situation, the asymmetry of the pressure

distribution within the droplet and at the solid–liquid contact surface was more significant, and the gradient-direction jumping velocity is greater. In addition, it should be noted that not all droplet center-of-mass lines were parallel or perpendicular to the trapezoidal bottom edge of the wedge-shaped micropillar during the condensation process. As the angle between the droplet center-of-mass line and the trapezoidal bottom edge increased, the velocity in the gradient direction gradually increased and the vertical velocity varied slightly (Section S4, Supporting Information). Overall, the relative position of the droplet center-of-mass line perpendicular to the trapezoidal bottom edge of the wedge-shaped micropillar was more conducive to the droplet's motion in the gradient direction.

CONCLUSIONS

In this study, three types of superhydrophobic surfaces with annular wedge-shaped micropillar arrays were designed; the motion of condensate droplets was observed, and two primary dynamic behaviors of coalescence-induced droplet jumping on a single micropillar and cross micropillars were identified. In addition, variations of the number density and average radius of droplets with time on surfaces with differing sizes of wedge-shaped micropillars were obtained statistically. The results indicated that the droplet number density on the three surfaces decreased with condensation time, while the average droplet radius increased. The wedge-shaped micropillar size influenced the distributions of the droplet size, droplet number density, and coverage. The energy analysis of the coalescence-induced droplet jumping process on the wedge-shaped micropillar revealed that the relative size and relative position of the droplets and micropillars were the primary factors affecting the droplet jumping and the effects of these two factors on the jumping velocity and direction were investigated further by numerical simulations. The results indicated that the presence of the wedge-shaped micropillar caused the asymmetric distribution of droplet internal pressure and solid–liquid contact surface pressure, which generated an unbalanced force driving the droplet motion in the gradient direction of the wedge-shaped micropillar, causing the droplet to jump away from the surface with both vertical and gradient-direction velocities. The relative position of the droplet center-of-mass line perpendicular to the bottom edge of the trapezoidal shape of the wedge-shaped micropillar was more favorable for droplet transport. Moreover, as the droplet-micropillar relative size increased, the ability of the wedge-shaped micropillar surface to transport droplets in the gradient direction increased and then decreased. When the droplet-micropillar relative size was too large or too small, droplets could not be transported in the gradient direction, which was also a limitation of transporting droplets on superhydrophobic surfaces with wedge-shaped micropillars. The investigation of coalescence-induced droplet jumping on superhydrophobic surfaces with annular wedge-shaped micropillar arrays further elucidates the influence of the surface structure on the jumping velocity and direction. The findings can guide the design of microstructures on superhydrophobic surfaces, which is crucial for applying droplet jumping in fields such as water harvesting, thermal management of electronic devices, and microfluidics.

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/acs.langmuir.3c02534>.

Energy analysis of coalescence-induced droplet jumping, the relative size of the droplet radius to the bottom edge of the wedge-shaped micropillars in the condensation experiment, the coalescence-induced droplet jumping on the superhydrophobic surface with no wedge-shaped micropillars, and the effect of the angle between the droplet center-of-mass line and the trapezoidal bottom edge of the wedge-shaped micropillar (PDF)

Dynamic behaviors of condensate droplets on the superhydrophobic surface with annular wedge-shaped micropillar arrays (MP4)

■ AUTHOR INFORMATION

Corresponding Authors

Xiaomin Wu – Key Laboratory for Thermal Science and Power Engineering of Ministry of Education, Department of Energy and Power Engineering, Tsinghua University, Beijing 100084, China; orcid.org/0000-0001-7703-0038; Email: wuxiaomin@mail.tsinghua.edu.cn

Zhiping Yuan – Department of Energy and Power Engineering, School of Mechanical Engineering, Beijing Institute of Technology, Beijing 100081, China; orcid.org/0000-0003-1601-9593; Email: nikolatesal@bit.edu.cn

Authors

Huimin Hou – Key Laboratory for Thermal Science and Power Engineering of Ministry of Education, Department of Energy and Power Engineering, Tsinghua University, Beijing 100084, China; orcid.org/0000-0003-4075-9313

Zhifeng Hu – Research Center of Solar Power and Refrigeration, School of Mechanical Engineering, Shanghai Jiao Tong University, Shanghai 200240, China

Sihaogao Gao – Key Laboratory for Thermal Science and Power Engineering of Ministry of Education, Department of Energy and Power Engineering, Tsinghua University, Beijing 100084, China

Complete contact information is available at:

<https://pubs.acs.org/doi/10.1021/acs.langmuir.3c02534>

Author Contributions

H. Hou and Z. Yuan proposed the idea. H. Hou designed the research and conducted the experiments. H. Hou derived the theory with the help of X. M. Wu. H. Hou wrote the initial manuscript. X. M. Wu, Z. Yuan, Z. Hu, and S. Gao reviewed and edited the manuscript.

Notes

The authors declare no competing financial interest.

■ ACKNOWLEDGMENTS

This work was supported by the National Natural Science Foundation of China (nos. 51976098 and 12202235) and the China Postdoctoral Science Special Foundation (no. 2023TQ0210).

■ REFERENCES

- (1) Liu, C.; Zhao, M.; Lu, D.; Sun, Y.; Song, L.; Zheng, Y. Laplace Pressure Difference Enhances Droplet Coalescence Jumping on Superhydrophobic Structures. *Langmuir* **2022**, *38*, 6923–6933.
- (2) Yan, X.; Chen, F.; Zhao, C.; Wang, X.; Li, L.; Khodakarami, S.; Fazle Rabbi, K.; Li, J.; Hoque, M. J.; Chen, F.; Feng, J.; Miljkovic, N. Microscale Confinement and Wetting Contrast Enable Enhanced and Tunable Condensation. *ACS Nano* **2022**, *16*, 9510–9522.
- (3) Ma, C.; Chen, L.; Wang, L.; Tong, W.; Chu, C.; Yuan, Z.; Lv, C.; Zheng, Q. Condensation droplet sieve. *Nat. Commun.* **2022**, *13*, 5381.
- (4) Hou, H.; Wu, X.; Hu, Z.; Gao, S.; Wu, Y.; Lin, Y.; Dai, L.; Zou, G.; Liu, L.; Yuan, Z. High-speed directional transport of condensate droplets on superhydrophobic saw-tooth surfaces. *J. Colloid Interface Sci.* **2023**, *649*, 290–301.
- (5) Wang, X.; Zeng, J.; Li, J.; Yu, X.; Wang, Z.; Zhang, Y. Beetle and cactus-inspired surface endows continuous and directional droplet jumping for efficient water harvesting. *J. Mater. Chem. A* **2021**, *9*, 1507–1516.
- (6) Sun, K.; Liu, Z.; Wei, J.; Wang, T. Thermal performance of a vapor chamber with synergistic effects of droplet jumping and pillared-wick capillarity. *Int. J. Heat Mass Transfer* **2022**, *195*, 123167.
- (7) Oh, J.; Birbarah, P.; Foulkes, T.; Yin, S. L.; Rentauskas, M.; Neely, J.; Pilawa-Podgurski, R. C. N.; Miljkovic, N. Jumping-droplet electronics hot-spot cooling. *Appl. Phys. Lett.* **2017**, *110*, 123107.
- (8) Boreyko, J. B.; Chen, C.-H. Vapor chambers with jumping-drop liquid return from superhydrophobic condensers. *Int. J. Heat Mass Transfer* **2013**, *61*, 409–418.
- (9) Cira, N. J.; Benusiglio, A.; Prakash, M. Vapour-mediated sensing and motility in two-component droplets. *Nature* **2015**, *519*, 446–450.
- (10) Wang, L.; Wu, H.; Wang, F. Efficient transport of droplet sandwiched between saw-tooth plates. *J. Colloid Interface Sci.* **2016**, *462*, 280–287.
- (11) Zheng, Y.; Cheng, J.; Zhou, C.; Xing, H.; Wen, X.; Pi, P.; Xu, S. Droplet Motion on a Shape Gradient Surface. *Langmuir* **2017**, *33*, 4172–4177.
- (12) Boreyko, J. B.; Chen, C. H. Self-propelled dropwise condensate on superhydrophobic surfaces. *Phys. Rev. Lett.* **2009**, *103*, 184501.
- (13) Wang, X.; Chen, Z.; Xu, B. Coalescence-induced jumping of condensate droplets on microstructured surfaces with different gravitational fields by lattice Boltzmann method. *Comput. Fluids* **2019**, *188*, 60–69.
- (14) Peng, Q.; Yan, X.; Li, J.; Li, L.; Cha, H.; Ding, Y.; Dang, C.; Jia, L.; Miljkovic, N. Breaking Droplet Jumping Energy Conversion Limits with Superhydrophobic Microgrooves. *Langmuir* **2020**, *36*, 9510–9522.
- (15) Gao, S.; Hu, Z.; Yuan, Z.; Wu, X. Flexible and efficient regulation of coalescence-induced droplet jumping on superhydrophobic surfaces with string. *Appl. Phys. Lett.* **2021**, *118*, 191602.
- (16) Xie, F.-F.; Wang, D.-Q.; Yang, Y.-R.; Wang, X.-D.; Lee, D.-J. Coalescence-induced jumping and condensation of argon nanodroplets in the Cassie or the Wenzel state on nanopillar-arrayed surfaces. *Colloids Surf., A* **2021**, *628*, 127269.
- (17) Gao, S.; Hu, Z.; Wu, X. Enhanced horizontal mobility of a coalesced jumping droplet on superhydrophobic surfaces with an asymmetric ridge. *Phys. Fluids* **2022**, *34*, 122104.
- (18) Gao, Y.; Ke, Z.; Yang, W.; Wang, Z.; Zhang, Y.; Wu, W. Coalescence-Induced Droplet Jumping on Honeycomb Bionic Superhydrophobic Surfaces. *Langmuir* **2022**, *38*, 9981–9991.
- (19) Vahabi, H.; Wang, W.; Mabry, J. M.; Kota, A. K. Coalescence-induced jumping of droplets on superomniphobic surfaces with macrotexture. *Sci. Adv.* **2018**, *4*, No. eaau3488.
- (20) Yuan, Z.; Hou, H.; Dai, L.; Wu, X.; Tryggvason, G. Controlling the Jumping Angle of Coalescing Droplets Using Surface Structures. *ACS Appl. Mater. Interfaces* **2020**, *12*, 52221–52228.
- (21) Lu, D.; Zhao, M.; Zhang, H.; Yang, Y.; Zheng, Y. Self-Enhancement of Coalescence-Induced Droplet Jumping on Superhydrophobic Surfaces with an Asymmetric V-Groove. *Langmuir* **2020**, *36*, 5444–5453.
- (22) Yin, C.; Wang, T.; Che, Z.; Jia, M.; Sun, K. Critical and Optimal Wall Conditions for Coalescence-Induced Droplet Jumping on Textured Superhydrophobic Surfaces. *Langmuir* **2019**, *35*, 16201–16209.

- (23) Feng, S.; Delannoy, J.; Malod, A.; Zheng, H.; Quéré, D.; Wang, Z. Tip-induced flipping of droplets on Janus pillars: From local reconfiguration to global transport. *Sci. Adv.* **2020**, *6*, No. eabb4540.
- (24) Kumar, M.; Bhardwaj, R.; Sahu, K. C. Wetting Dynamics of a Water Droplet on Micropillar Surfaces with Radially Varying Pitches. *Langmuir* **2020**, *36*, 5312–5323.
- (25) Lecointre, P.; Laney, S.; Michalska, M.; Li, T.; Tanguy, A.; Papakonstantinou, I.; Quéré, D. Unique and universal dew-repellency of nanocones. *Nat. Commun.* **2021**, *12*, 3458.
- (26) Zhu, Y.; Yan, X.; Ma, L.; Liu, Z.; Liu, W. Exploring the Role of Initial Droplet Position in Coalescence-Induced Droplet Jumping: Lattice Boltzmann Simulations. *Processes* **2022**, *10*, 986.
- (27) Mulroe, M. D.; Srijanto, B. R.; Ahmadi, S. F.; Collier, C. P.; Boreyko, J. B. Tuning Superhydrophobic Nanostructures To Enhance Jumping-Droplet Condensation. *ACS Nano* **2017**, *11*, 8499–8510.
- (28) Attarzadeh, R.; Dolatabadi, A. Coalescence-induced jumping of micro-droplets on heterogeneous superhydrophobic surfaces. *Phys. Fluids* **2017**, *29*, 012104.
- (29) Shi, Y.; Tang, G. H.; Xia, H. H. Investigation of coalescence-induced droplet jumping on superhydrophobic surfaces and liquid condensate adhesion on slit and plain fins. *Int. J. Heat Mass Transfer* **2015**, *88*, 445–455.
- (30) Liu, J.; Guo, H.; Zhang, B.; Qiao, S.; Shao, M.; Zhang, X.; Feng, X. Q.; Li, Q.; Song, Y.; Jiang, L.; Wang, J. Guided Self-Propelled Leaping of Droplets on a Micro-Anisotropic Superhydrophobic Surface. *Angew. Chem., Int. Ed. Engl.* **2016**, *55*, 4265–4269.
- (31) Wang, X.; Zeng, J.; Yu, X.; Liang, C.; Zhang, Y. Water harvesting method via a hybrid superwetable coating with superhydrophobic and superhydrophilic nanoparticles. *Appl. Surf. Sci.* **2019**, *465*, 986–994.
- (32) Li, D.; Qian, C.; Gao, S.; Zhao, X.; Zhou, Y. Self-propelled drop jumping during defrosting and drainage characteristic of frost melt water from inclined superhydrophobic surface. *Int. J. Refrig.* **2017**, *79*, 25–38.
- (33) Zhang, Q.; Lin, G.; Yin, J. Highly efficient fog harvesting on superhydrophobic microfibers through droplet oscillation and sweeping. *Soft Matter* **2018**, *14*, 8276–8283.
- (34) Hou, H.; Wu, X.; Hu, Z.; Gao, S.; Dai, L.; Yuan, Z. Coalescence-induced droplet jumping on superhydrophobic surfaces with non-uniformly distributed micropillars. *Colloids Surf., A* **2023**, *675*, 132127.
- (35) Hou, H.; Yuan, Z.; Hu, Z.; Gao, S.; Wu, X. Effects of the surface tension gradient and viscosity on coalescence-induced droplet jumping on superamphiphobic surfaces. *Phys. Fluids* **2021**, *33*, 112101.
- (36) Yuan, Z.; Wu, X.; Chu, F.; Wu, R. Numerical simulations of guided self-propelled jumping of droplets on a wettability gradient surface. *Appl. Therm. Eng.* **2019**, *156*, 524–530.